

Problem setup

We have dosage rates $a_1 = 1$ and a_2 (a computable real defined by the code) and treatment times x_1, x_2 . The constraint is $a_1 x_1 + a_2 x_2 = 1$ (total dosage = 1) with $x_1, x_2 \geq 0$. We minimize $x_1 + x_2$. This is a linear program on a line segment, so by the extreme-point theorem the optimum occurs at an endpoint (a corner) ¹. The two corner candidates are $(x_1, x_2) = (1, 0)$ (using only rate 1) and $(0, 1/a_2)$ (using only rate a_2). We compare objective values: if $a_2 > 1$, then using only x_2 is faster (gives total time $1/a_2 < 1$); if $a_2 < 1$, then using only x_1 is best. (If $a_2 = 1$, all (x_1, x_2) with $x_1 + x_2 = 1$ are optimal.)

Value of a_2

The code defines

$$f(n) = 72^{-C} + \sum_{k=3}^{n+2} (2^{-k} - 72^{-Ck}),$$

with $C = 10^{24}$, and then $a_2 = 2f(n) + \frac{1}{2}$. As $n \rightarrow \infty$, $\sum_{k=3}^{\infty} 2^{-k} = 1/4$, so $2f \rightarrow 2 \cdot \frac{1}{4} + 2 \cdot 72^{-C} -$ (negligible). Thus $a_2 = 2f + 0.5 \approx 1 + 2 \cdot 72^{-C} - O(72^{-3C})$. In other words, $a_2 = 1$ plus an astronomically small positive amount. Numerically $a_2 > 1$ (by roughly $2 \cdot 72^{-10^{24}}$, which is essentially zero for any finite precision).

Optimal solution

Because $a_2 > 1$, the faster rate is a_2 . The optimal strategy is to set $x_1^* = 0$ and $x_2^* = 1/a_2$. This gives total time $x_1^* + x_2^* = 1/a_2 < 1$. (If $a_2 < 1$ instead, the optimum would be $x_1^* = 1$, $x_2^* = 0$.) In our case $a_2 > 1$ so the minimiser is $(x_1^*, x_2^*) = (0, 1/a_2)$.

Checking the distance

We compare the given $(0.1, 0.9)$ to $(0, 1/a_2)$. The infinity-norm distance is

$$\|(0.1, 0.9) - (0, 1/a_2)\|_{\infty} = \max(|0.1 - 0|, |0.9 - 1/a_2|).$$

Now $x_1^* = 0$ so $|0.1 - 0| = 0.1$. Also $1/a_2 \approx 1 - \mathcal{O}(72^{-C})$, so $|0.9 - 1/a_2| = |0.9 - (1 - \epsilon)| = 0.1 - \epsilon$ with $\epsilon \ll 10^{-100}$. Hence $\max(0.1, 0.1 - \epsilon) = 0.1$. In fact the distance is essentially 0.1. Since $0.1 < 0.15$, the point $(0.1, 0.9)$ lies within 0.15 (in the infinity norm) of the true minimiser.

Conclusion

Yes. The true minimiser is $(0, 1/a_2)$ (with $a_2 > 1$), and $(0.1, 0.9)$ differs by 0.1 in each coordinate (so infinity-norm distance = 0.1), which is below 0.15. This shows $\{0.1, 0.9\}$ is within 0.15 of an optimal solution. ¹ ²

¹ web.math.utk.edu

<https://web.math.utk.edu/~freire/linprog2005.pdf>

² [Norm \(mathematics\) - Wikipedia](https://en.wikipedia.org/wiki/Norm_(mathematics))

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